

ADVANCED STATISTICAL MECHANICS

- 1.1. (a) Show that, for two *large* systems in thermal contact, the number $\Omega^{(0)}(E^{(0)}, E_1)$ of Section 1.2 can be expressed as a Gaussian in the variable E_1 . Determine the root-mean-square deviation of E_1 from the mean value $\bar{E}_1$ in terms of other quantities pertaining to the problem.
- (b) Make an explicit evaluation of the root-mean-square deviation of E_1 in the special case when the systems A_1 and A_2 are ideal classical gases.

Let us set up the problem &

- We have systems A_1 & A_2
- Total Energy & Total energy is constant $E^\circ = E_1 + E_2$
- # of microstates Ω & For a composite system A° & $\Omega^\circ(E^\circ, E_1) = \Omega_1(E_1) \Omega_2(E_2)$
 $= \Omega_1(E_1) \Omega_2(E^\circ - E_1)$
- Entropy S° & Total entropy of A° & $S^\circ(E_1) = k \ln[\Omega^\circ(E^\circ, E_1)]$
 $= S_1(E_1) + S_2(E_2)$ OR $S_1(E_1) + S_2(E^\circ - E_1)$

a) Goal & Show that $\Omega^\circ \sim$ Gauss in E_1 & find RMS.

proof

We know for 2 large systems, equilibrium value $\bar{E}_1$ maximises $S(E_1)$ such that

$$\left. \frac{dS(E_1)}{dE_1} \right|_{\bar{E}_1} = 0$$

Taylor Expanding $S(E_1)$ about $\bar{E}_1$ &

$$S(E_1) = S(\bar{E}_1) + \cancel{\left. \frac{dS(E_1)}{dE_1} \right|_{\bar{E}_1} (E_1 - \bar{E}_1)} + \frac{1}{2} \left. \frac{d^2 S(E_1)}{dE_1^2} \right|_{\bar{E}_1} (E_1 - \bar{E}_1)^2 + \dots$$

Rewriting this using Ω° &

$$S(E_1) = k \ln[\Omega^\circ(E^\circ, E_1)] \approx \underbrace{S(\bar{E}_1)}_{\text{constant}} + \frac{1}{2} \left. \frac{d^2 S(E_1)}{dE_1^2} \right|_{\bar{E}_1} (E_1 - \bar{E}_1)^2$$

$$\Rightarrow \Omega^\circ(E^\circ, E_1) \approx \underbrace{\exp\left\{\frac{S(\bar{E}_1)}{k}\right\}}_{\text{constant}} \cdot \exp\left\{\frac{1}{2k} \left. \frac{d^2 S}{dE_1^2} \right|_{\bar{E}_1} (E_1 - \bar{E}_1)^2\right\}$$

$$\Omega^\circ(E^\circ, E_1) \propto \exp\left\{-\frac{(E_1 - \bar{E}_1)^2}{2\sigma^2}\right\} \quad \text{with } \sigma^2 = \frac{k}{\left. \frac{d^2 S}{dE_1^2} \right|_{\bar{E}_1}}$$

↑
ignoring
higher order
terms

We showed that Ω^0 is gaussian in E_1 . We now continue to find RMS.

$$RMS = \sqrt{\langle (E_1 - \bar{E}_1)^2 \rangle} = \sqrt{\sigma^2} \quad \text{where} \quad \sigma^2 = \frac{k}{d^2S/dE_1^2} \Big|_{\bar{E}_1}$$

Evaluating $\frac{d^2S}{dE_1^2} \Big|_{\bar{E}_1}$

$$\begin{aligned} \frac{dS}{dE_1} &= \frac{d(S_1 + S_2)}{dE_1} = \underbrace{\frac{dS_1}{dE_1}}^{\frac{1}{T_1}} + \underbrace{\frac{dS_2}{dE_2}}^{\frac{1}{T_2}} \cdot \underbrace{\frac{dE_2}{dE_1}}^{-1} \\ &= \frac{1}{T_1} - \frac{1}{T_2} \end{aligned}$$

$$\begin{aligned} \frac{d^2S}{dE_1^2} &= \frac{d}{dE_1} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) = -\frac{1}{T_1^2} \cdot \underbrace{\frac{dT_1}{dE_1}}^{\frac{1}{C_1}} + \frac{1}{T_2^2} \cdot \underbrace{\frac{dT_2}{dE_2}}^{\frac{1}{C_2}} \cdot \underbrace{\frac{dE_2}{dE_1}}^{-1} \\ &= -\frac{1}{T_1^2 C_1} - \frac{1}{T_2^2 C_2} \end{aligned}$$

At equilibrium, $T_1 = T_2 = T$. Therefore,

$$\frac{d^2S}{dE_1^2} \Big|_{\bar{E}_1} = -\frac{1}{T^2} \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

We finally have that

$$RMS = \sqrt{\sigma^2} = \sqrt{\frac{KT^2}{\frac{1}{C_1} + \frac{1}{C_2}}} = \sqrt{KT^2 \frac{C_1 C_2}{C_1 + C_2}}$$

b) For ideal gases $C_i = \frac{3}{2} N_i k$

$$\Rightarrow RMS = \sqrt{\frac{3}{2} k^2 T^2 \frac{N_1 N_2}{N_1 + N_2}}$$

1.2. Assuming that the entropy S and the statistical number Ω of a physical system are related through an arbitrary functional form

$$S = f(\Omega),$$

show that the additive character of S and the multiplicative character of Ω necessarily require that the function $f(\Omega)$ be of the form (1.2.6).

Goal : Show that $S = k \ln \Omega$ using the additive character of S and the multiplicative character of Ω

proof | Let A_1 & A_2 be system with entropy & number of microstates S_1, S_2 & Ω_1, Ω_2 respectively. Then,

$$S = S_1 + S_2 \quad \text{additive prop}$$

$$f(\Omega_1 \Omega_2) = f(\Omega_1) + f(\Omega_2) \quad \text{multiplicative prop}$$

Taking the derivative in terms of Ω_1 & Ω_2 , we get

$$\Omega_2 \frac{\partial f(\Omega_1, \Omega_2)}{\partial \Omega_1} = \frac{\partial f(\Omega_1)}{\partial \Omega_1} \quad \& \quad \Omega_1 \frac{\partial f(\Omega_1, \Omega_2)}{\partial \Omega_2} = \frac{\partial f(\Omega_2)}{\partial \Omega_2}$$

$$\Omega_1 \frac{\partial f(\Omega_1)}{\partial \Omega_1} = \Omega_1 \Omega_2 \frac{\partial f(\Omega_1, \Omega_2)}{\partial \Omega_1} = \Omega_2 \Omega_1 \frac{\partial f(\Omega_1, \Omega_2)}{\partial \Omega_2} = \Omega_2 \frac{\partial f(\Omega_2)}{\partial \Omega_2}$$

Therefore, for any Ω & Ω $f'(\Omega) = k$ constant

$$\Rightarrow f'(\Omega) = \frac{k}{\Omega} \xrightarrow[\ln \Omega]{\text{integrating}} f(\Omega) = k \ln \Omega + C$$

If we have 1 microstate, then the entropy must be 0 implying that

$$S = f(1) = k \ln 1 + C = 0 \quad \text{so} \quad C = 0$$

$$\therefore S = k \ln \Omega \quad \square$$

- 1.3. Two systems, A and B, of identical composition, are brought together and allowed to exchange both energy and particles, keeping volumes V_A and V_B constant. Show that the minimum value of the quantity (dE_A/dN_A) is given by

$$\frac{\mu_A T_B - \mu_B T_A}{T_B - T_A},$$

where the μ 's and the T 's are the respective chemical potentials and temperatures.

Let two systems A & B of identical composition exchange E, N but V_A & V_B constant

Then, the differential entropy is s (fixed V)

$$dE_i = T_i dS_i - \cancel{P_i dV_i} + \mu_i dN_i$$

$$dS_i = \frac{dE_i}{T_i} - \frac{\mu_i dN_i}{T_i}$$

We have for this system the following constraints :

$\Rightarrow$ Change in Entropy is positive & $dS_{\text{Total}} \geq 0$ for $S_{\text{Total}} = S_A + S_B$

$\Rightarrow$ Total Energy & number of particles constant : $E_A + E_B = E_{\text{Total}}$ $N_A + N_B = N_{\text{Total}}$
 $dE_A + dE_B = 0$ $dN_A + dN_B = 0$

We now evaluate dS_{Total} to get $\frac{dE_A}{dN_A}$!

$$\begin{aligned} dS_{\text{Total}} &= dS_A + dS_B \\ &= \frac{dE_A}{T_A} - \frac{\mu_A dN_A}{T_A} + \frac{dE_B}{T_B} - \frac{\mu_B dN_B}{T_B} \\ &= \frac{dE_A}{T_A} - \frac{\mu_A dN_A}{T_A} + \frac{-dE_A}{T_B} - \frac{\mu_B (-dN_A)}{T_B} \quad \text{from constraint (2)} \\ &= \left(\frac{1}{T_A} - \frac{1}{T_B} \right) dE_A + \left(\frac{\mu_B}{T_B} - \frac{\mu_A}{T_A} \right) dN_A \end{aligned}$$

At equilibrium, $dS_{\text{Total}} = 0$ & $\frac{dE_A}{dN_A}$ is minimum, $\therefore$

$$\left(\frac{1}{T_A} - \frac{1}{T_B} \right) dE_A + \left(\frac{\mu_B}{T_B} - \frac{\mu_A}{T_A} \right) dN_A = 0$$

$$\frac{dE_A}{dN_A} \Big|_{\text{min}} = \frac{\left(\frac{\mu_A}{T_A} - \frac{\mu_B}{T_B} \right)}{\left(\frac{1}{T_A} - \frac{1}{T_B} \right)} = \frac{\mu_A T_B - \mu_B T_A}{T_B - T_A}$$



1.4. In a classical gas of hard spheres (of diameter D), the spatial distribution of the particles is no longer uncorrelated. Roughly speaking, the presence of j particles in the system leaves only a volume $(V - jv_0)$ available for the $(j+1)$ st particle; clearly, v_0 would be proportional to D^3 . Assuming that $Nv_0 \ll V$, determine the dependence of $\Omega(N, V, E)$ on V (compare with equation (1.4.1)) and show that, as a result of this, V in the ideal-gas law (1.4.3) gets replaced by $(V - b)$, where b is four times the actual volume occupied by the particles.

Goals For a classical gas of hard spheres,

- find f s.t. $\Omega(N, V, E) \propto f(V)$
- New ideal gas law is $P(V-b) = NkT$

proof

For an ideal gas, the # of microstates Ω is:

$$\Omega(N, V, E) \propto V^N \quad 1.4.1$$

However, now for hard spheres, particles will occupy volume v_0 that no other particle can occupy. Therefore,

$$1^{\text{st}} \text{ particle} \rightarrow V$$

$$2^{\text{nd}} \text{ particle} \rightarrow V - v_0$$

$$3^{\text{rd}} \text{ particle} \rightarrow V - 2v_0$$

$\vdots$

$$n^{\text{th}} \text{ particle} \rightarrow V - (n-1)v_0$$

Therefore, the # of microstates for the new set up will be proportional to:

$$\Omega(N, V, E) \propto V(V-v_0)(V-2v_0)\dots(V-(N-1)v_0)$$

We can now use the asymptotic relation $Nv_0 \ll V$ to simplify

$$\begin{aligned} \ln \Omega(N, V, E) &= \sum_{i=0}^{N-1} \ln[V - iv_0] \\ &= \sum_{i=0}^{N-1} \left(\ln[V] + \ln\left[1 - \frac{iv_0}{V}\right] \right) \\ &\approx N \ln V + \sum_{i=0}^{N-1} -\frac{iv_0}{V} \\ &= N \ln V - \frac{N(N-1)}{2} \frac{v_0}{V} \\ &\approx N \ln V - \frac{N^2}{2} \frac{v_0}{V} \end{aligned}$$

$$\text{Taylor Expansion: } \frac{iv_0}{V} < \frac{Nv_0}{V} \ll 1 \Rightarrow \ln(1-x) = -x - \dots$$

$$\sum_{i=0}^{N-1} i = 0+1+\dots+N-1 = \frac{(N-1)N}{2}$$

Going back now and exponentiating we get

$$\Omega(N, V, E) \approx V^N e^{-\frac{N^2 v_0}{2V}} \approx V^N \left(1 - \frac{N^2 v_0}{2V}\right)$$

Taylor expansion: $Nv_0 \ll V$

$$e^{-x} = 1 - x + \dots$$

Goal 1 achieved!

As for goal 2, we use some thermodynamic relationships for ideal gases:

$$S = k \ln \Omega, \quad PV = NkT \quad \& \quad \frac{P}{T} = \left(\frac{\partial S}{\partial V} \right)_{N, E} \quad (1.4.2)$$

$$\frac{P}{T} = \frac{\partial (k \ln \Omega)}{\partial V} = k \frac{\partial}{\partial V} \left[N \ln V - \frac{N^2 v_0}{2V} \right]$$

$$= k \left[\frac{N}{V} + \frac{N^2 v_0}{2V^2} \right]$$

$$= \frac{N}{V} k \left[1 + \frac{Nv_0}{2V} \right]$$

Rearranging we get $P V \left[1 + \frac{Nv_0}{2V} \right]^{-1} = NkT \leftarrow$ our new ideal gas law

Again since $Nv_0 \ll V \Rightarrow PV \left(1 - \frac{Nv_0}{2V} \right) = NkT$

Geometric series sum: $\frac{1}{1+x} = 1 - x + x^2 - \dots$

Therefore, our V_{eff} takes the form $V_{\text{eff}} = V - \frac{Nv_0}{2}$ or $V_{\text{eff}} = V - b$

What is the physical meaning of $b = \frac{Nv_0}{2}$?

Figure 1

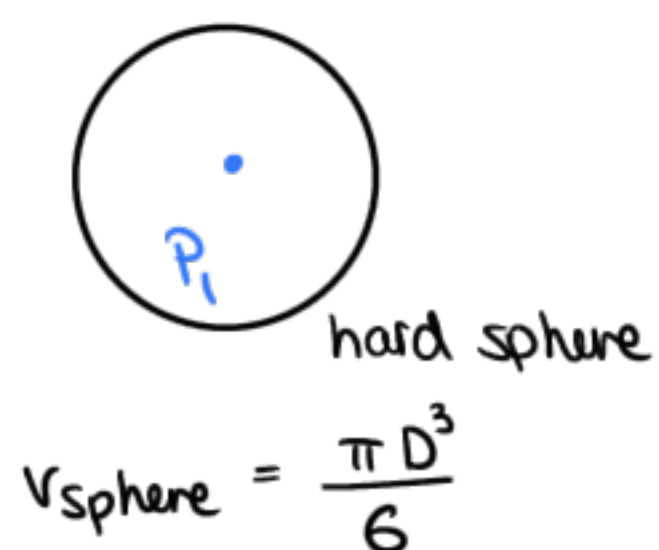
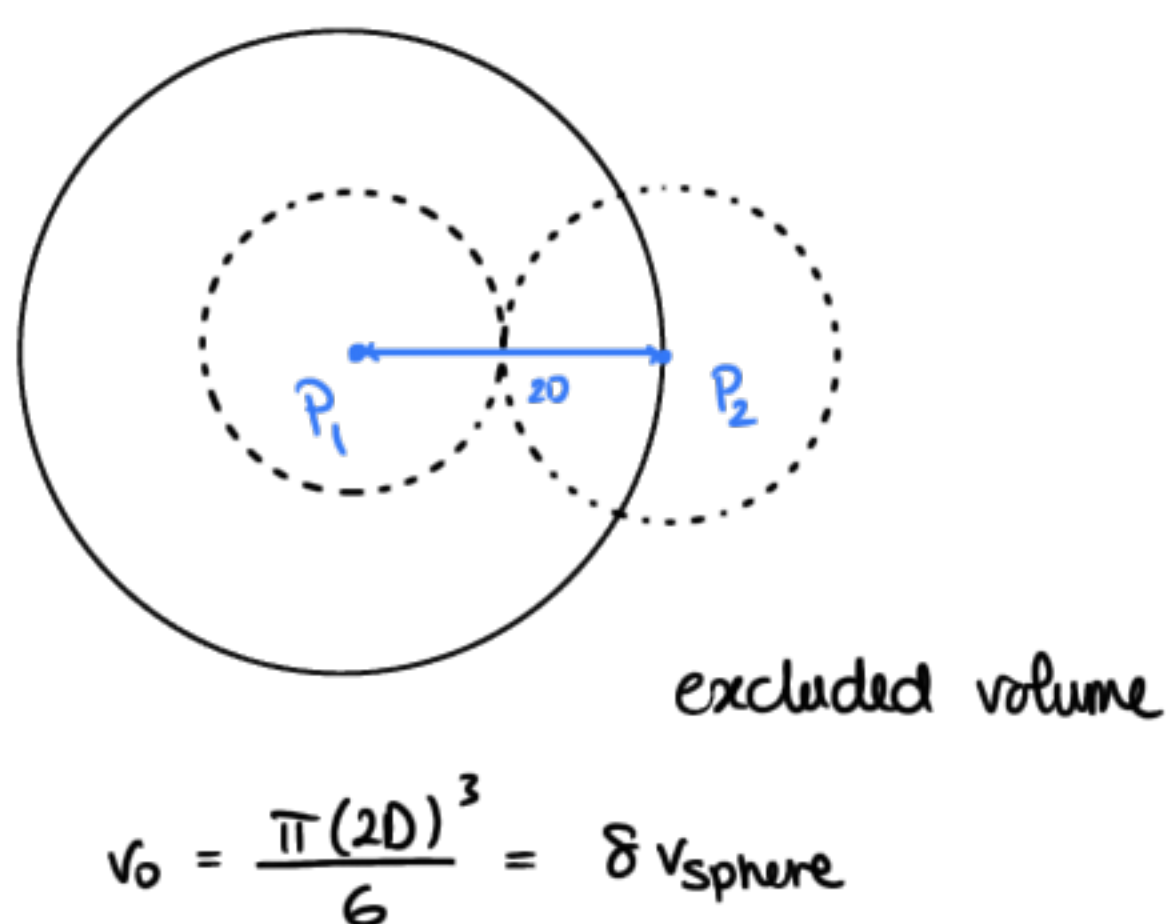


Figure 2



$$b = \frac{Nv_0}{2} = \frac{N(8v_{\text{sphere}})}{2} = 4Nv_{\text{sphere}}$$

OR 4 times the actual volume of particles

Centers of particles can't lie on boundaries of the hard spheres. So the excluded volume is where it is permissible to put another particles centre (Figure 2)